



National University of Computer & Emerging Sciences, Karachi

Fall -2025 CS/CY/SE-Department

Assignment- 2

18th September 2025



Course Code: CS2009	Course Name: Design & Analysis of Algorithms
Instructor Name:	
Date of Submission	30 th September 2025

Instructions:

Max Marks: 55

- Make your Assignments on Full Scale Papers, Typed Assignments will not be acceptable:
- 20% penalty for 1 day late
- 40% penalty for 2 days late
- Submission not allowed afterwards

1) A large e-commerce company needs to process a massive list of orders based on their total value. The orders are represented as integers, and the company wants to prioritize the orders with the highest value. Design an efficient algorithm to sort the orders in descending order of their total value using Quick Sort (Divide and Conquer) and compute the time complexity of your designed algorithm. Analyse and state the **time complexity** of your algorithm (best, average, and worst case).

(10 Marks)

2) A university's registrar office needs to process a list of student scores for a particular exam. The list of scores is as follows: [59,6, 35, 12, 27, 9, 1, 18, 5, 31,16]. To facilitate grading and feedback, the office wants to sort the list of scores in ascending order.

(10 Marks)

1. By applying Merge Sort generate a step-by-step algorithm to sort the list of student scores
2. Show the detailed steps of the Merge Sort algorithm to sort the list in ascending order.
3. Consider a variation of the Merge Sort algorithm that divides the input into four parts instead of two. Analyse the overall asymptotic running time of this modified algorithm. Write the recurrence relation for this modified algorithm.

3) A leading bank is developing a **financial risk analysis engine** to evaluate multiple investment portfolios under various market conditions. Each portfolio is represented by a **4×4 weight matrix (W)** indicating the distribution of funds across sectors such as technology, energy, healthcare, and finance, while the expected sector returns under different market conditions are stored in another **4×4 returns matrix (R)**. To compute the overall risk-return scores, the system must efficiently perform **matrix multiplication of $W \times R$** .

(10 Marks)

As part of this task, you are required to

- a. implement a **standard matrix multiplication algorithm** using the naïve triple-loop method and analyse its time complexity,

- b. implement the **Strassen's matrix multiplication algorithm** using the divide-and-conquer approach by splitting the 4×4 matrices into 2×2 submatrices and applying the seven Strassen formulas (M1–M7)
- c. Write the recurrence relation for **Strassen's matrix multiplication algorithm**
- d. Derive the time complexity by applying Master Theorem
- e. compare the **time complexities** of both methods

4) Discuss all cases of Master Theorem. (5 Marks)

5) Using the Master Theorem, can we solve the modified recurrence relation of question. (5 Marks)

6) Use Iteration Method and Recurrence tree, to calculate the time complexity of the following (7.5 Marks)

- I. $T(n) = 2T(n/2) + n^2$
- II. $T(n) = 3T(n/3) + n \log^2 n$
- III. $T(n) = 2T(n/2) + \log n$

7) Use Master Theorem, to calculate the time complexity of the following (7.5 Marks)

$$T(n) = 4T(n/2) + n^2$$

- I. $T(n) = 2T(n/4) + \sqrt{n}$
- II. $T(n) = 4T(n/2) + n^2 \log n$

8) Apply the Substitution (Guess & Test) Method on the given recurrence relation to verify whether the provided guess is correct or not. (5 Marks)

I. $T(n) = 4T(n/2) + n^2$

Guess 1: $T(n) = O(n)$

Guess 2: $T(n) = O(n^2)$